BASE CHANGE FOR LOWER SHRIEK ALONG SEPARATED LOCALLY FINITE TYPE MORPHISMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Stalk description of lower shriek along separated morphisms (Lemma lemma-stalk-f-shriek-separated, canonical tag 0F76). cited at proof lines 796-805. Supplies, with its proof, the displayed stalk diagram whose arrow properties (isomorphisms, equalities, injectivities) produce the unique dotted bijection.
- Pullback maps on pushforwards (Sites, Section sites-section-pullback, canonical tag 06UM). cited at proof lines 777-779. Provides the general pullback map from which the base change map for lower shriek is obtained by restriction.
- Exactness of stalks (Etale Cohomology, Theorem etale-cohomology-theorem-exactness-stalks, canonical tag 03PU). cited at proof lines 825-828. Turns the stalk-level statements (membership in the subsheaf and bijectivity) into the sheaf-level isomorphism.

Source reference. Original theorem 03PU: sources/stacks-chow/tags.txt, source lines 4871–4871. Cited at proof lines 825-828: lifts the stalk-level membership and bijectivity to the sheaf-level isomorphism.

Source reference. Original lemma 0F76: sources/stacks-chow/tags.txt, source lines 18996–18996. Cited at proof lines 796-805: with its proof, supplies the displayed stalk diagram and the properties of its arrows.

Source reference. Original reference 06UM: sources/stacks-chow/tags.txt, source lines 8751–8751. Cited at proof lines 777-779: the general pullback map that restricts to the base change map for lower shriek.

Lemma 0.2. *Consider a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

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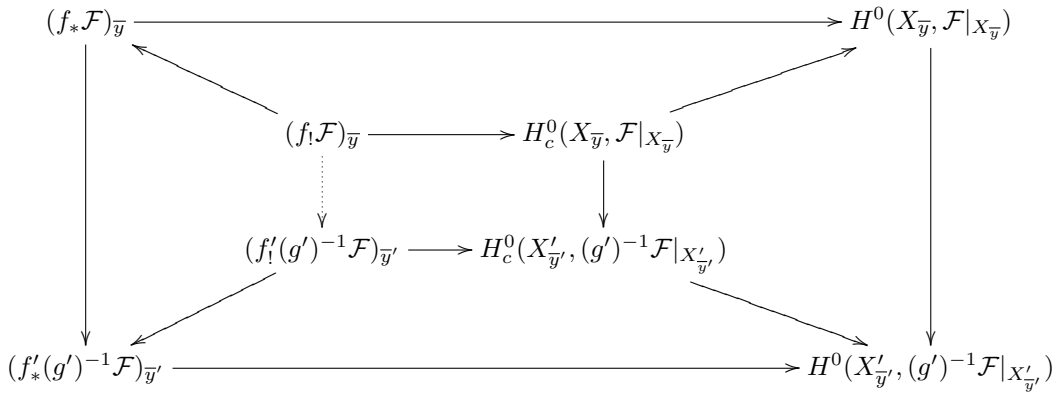
of schemes with f separated and locally of finite type. For any abelian sheaf $\mathcal{F}$ on $X_{étale}$ we have $f'_!(g')^{-1}\mathcal{F} = g^{-1}f_!\mathcal{F}$.

Proof. In great generality there is a pullback map $g^{-1}f_*\mathcal{F} \to f'_*(g')^{-1}\mathcal{F}$, see Sites, Section 06UM. We claim that this map sends $g^{-1}f_!\mathcal{F}$ into the subsheaf $f'_!(g')^{-1}\mathcal{F}$ and induces the isomorphism in the lemma.

Choose a geometric point $\overline{y}' : \mathrm{Spec}(k) \to Y'$ and denote $\overline{y} = g \circ \overline{y}'$ the image in Y. There is a commutative diagram

$$\begin{array}{ccc} (f_*\mathcal{F})_{\overline{y}} & \longrightarrow & H^0(X_{\overline{y}}, \mathcal{F}|_{X_{\overline{y}}}) \\ \downarrow & & \downarrow \\ (f'_*(g')^{-1}\mathcal{F})_{\overline{y}'} & \longrightarrow & H^0(X'_{\overline{y}'}, (g')^{-1}\mathcal{F}|_{X'_{\overline{y}'}}) \end{array}$$

where the horizontal maps were used in the proof of Lemma 0F76 and the vertical maps are the pullback maps above. The diagram commutes because each of the four maps in question is given by pulling back local sections along a morphism of schemes and the underlying diagram of morphisms of schemes commutes. Since the diagram in the statement of the lemma is cartesian we have $X'_{\overline{y}'} = X_{\overline{y}}$. Hence by Lemma 0F76 and its proof we obtain a commutative diagram



where the horizontal arrows of the inner square are isomorphisms and the two right vertical arrows are equalities. Also, the se, sw, ne, nw arrows are injective. It follows that there is a unique bijective dotted arrow fitting into the diagram. We conclude that $g^{-1}f_!\mathcal{F} \subset g^{-1}f_*\mathcal{F} \to f'_*(g')^{-1}\mathcal{F}$ is mapped into the subsheaf $f'_!(g')^{-1}\mathcal{F} \subset f'_*(g')^{-1}\mathcal{F}$ because this is true on stalks, see Étale Cohomology, Theorem 03PU. The same theorem then implies that the induced map is an isomorphism and the proof is complete. □